Natural Language Processing with Deep Learning

Farida Zehraoui
farida.zehraoui@univ-tours.fr
Polytech' Tours
Laboratoire d'informatique fondamentale et appliqué de Tours (LIFAT)
Université de Tours







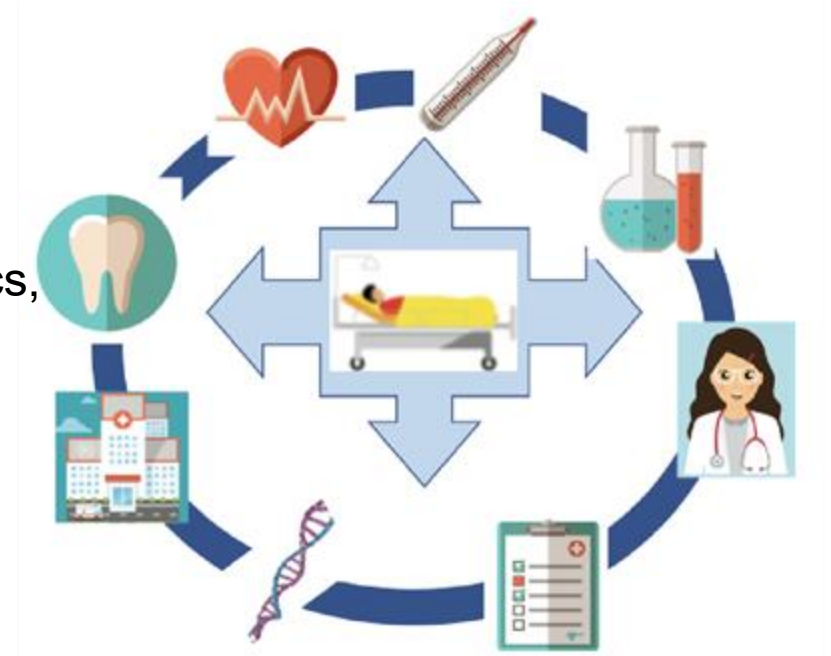
TP 4:
Medical Notes classification

Application context: Healthcare

Electronic Health Records (EHRs)

EHRs contain patients' medical history:

- ❑ Omics data: genomics, transcriptomics, proteomics, epigenomics, …
- ❑ Clinical data: medical notes, demographics, medical history, diagnoses, medications, laboratory test results, …
- ❑ Medical imaging: IRM, Scanners, Radiographies, …
- ❑ Signals: EEG, ECG, EMG, …
- ❑ …



Electronic Health Records (EHRs)

➡ Use of Artificial Intelligence: Machine Learning

Medical Applications

- Screening
- Diagnosis
- Therapy
- Prognosis
- …

Medical Applications

- **Screening**
- Diagnosis
- Therapy
- Prognosis
- …

Deep Learning for Screening COVID-19 using Chest X-Ray Images

Sanhita Basu
Department of Computer Science
West Bengal State University
West Bengal 700126, India
tania.sanhita@gmail.com

Sushmita Mitra
Machine Intelligence Unit
Indian Statistical Institute
Kolkata 700108, India
sushmita@isical.ac.in

Nilanjan Saha
Centre for Translational & Clinical Research
Jamia Hamdard (Deemed University),
New Delhi 110062, India
nilanjan.saha@jamiahamdard.ac.in

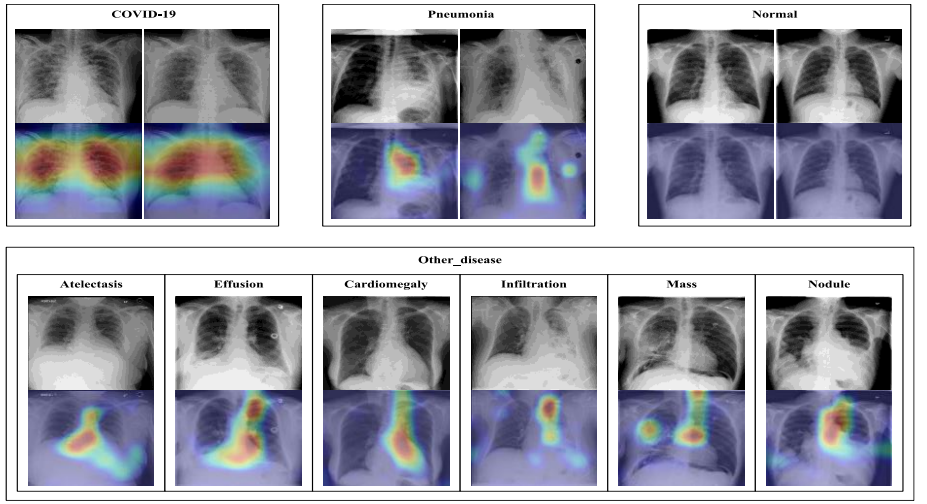


Fig. 4. Visual results obtained by Grad-CAM on different disease classes.

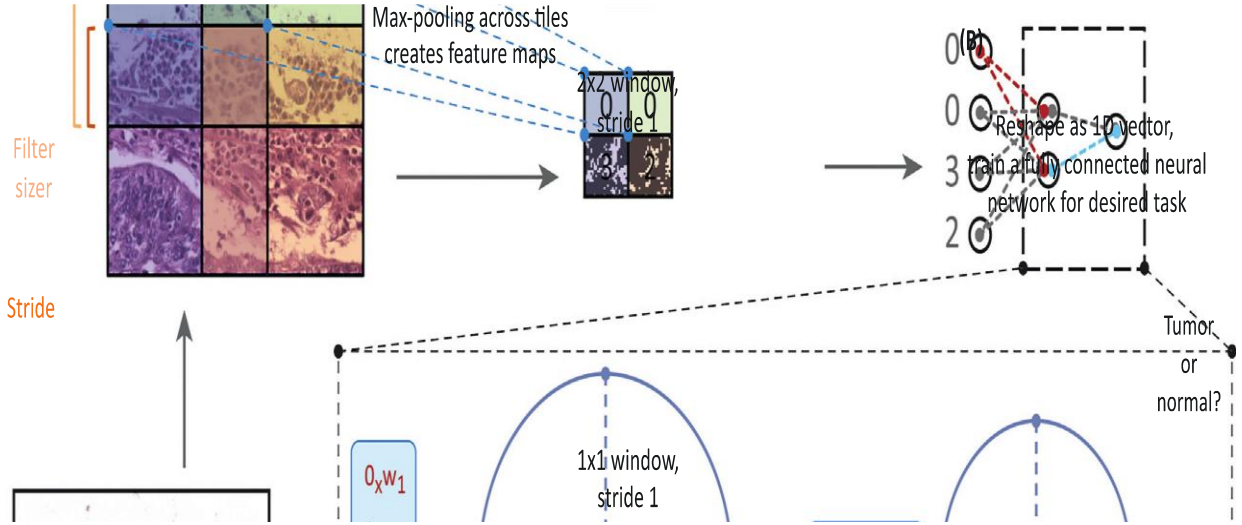
Medical Applications

- Screening
- **Diagnosis**
- Therapy
- Prognosis
- …

Review

Rise of the Machines: Advances in Deep Learning for Cancer Diagnosis

Adrian B. Levine,[1] Colin Schlosser,[2] Jasleen Grewal,[2] Robin Coope,[2] Steve J.M. Jones,[2] and Stephen Yip [1,*]



Medical Applications

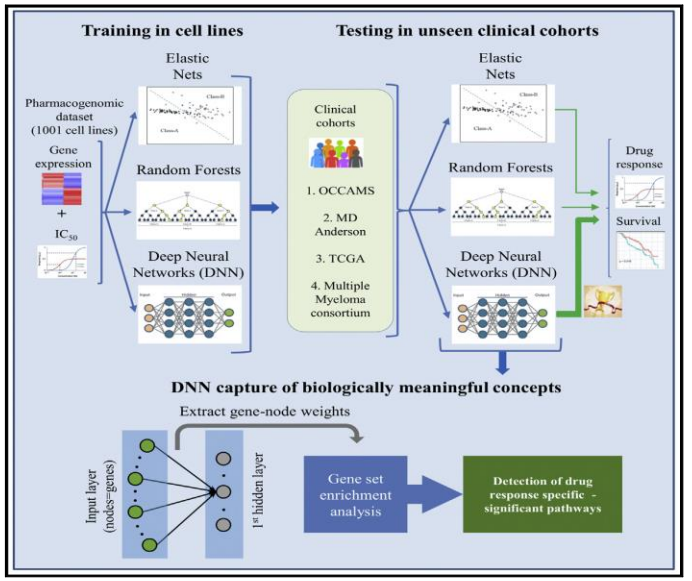
- Screening
- Diagnosis
- **Therapy**
- Prognosis
- …

Report

Cell Reports

A Deep Learning Framework for Predicting Response to Therapy in Cancer

Graphical Abstract



Authors

Theodore Sakellaropoulos, Konstantinos Vougas, Sonali Narang, ..., Russell Petty, Aristotelis Tsirigos, Vassilis G. Gorgoulis

Correspondence

kvougas@bioacademy.gr (K.V.), aristotelis.tsirigos@nyulangone.org (A.T.), vgorg@med.uoa.gr (V.G.G.)

In Brief

Sakellaropoulos et al. designed a machine learning workflow to predict drug response and survival of cancer patients. All pipelines are trained on a large panel of cancer cell lines and tested in clinical cohorts. DNN outperforms other machine learning algorithms by capturing pathways that link gene expression with drug response.

Medical Applications

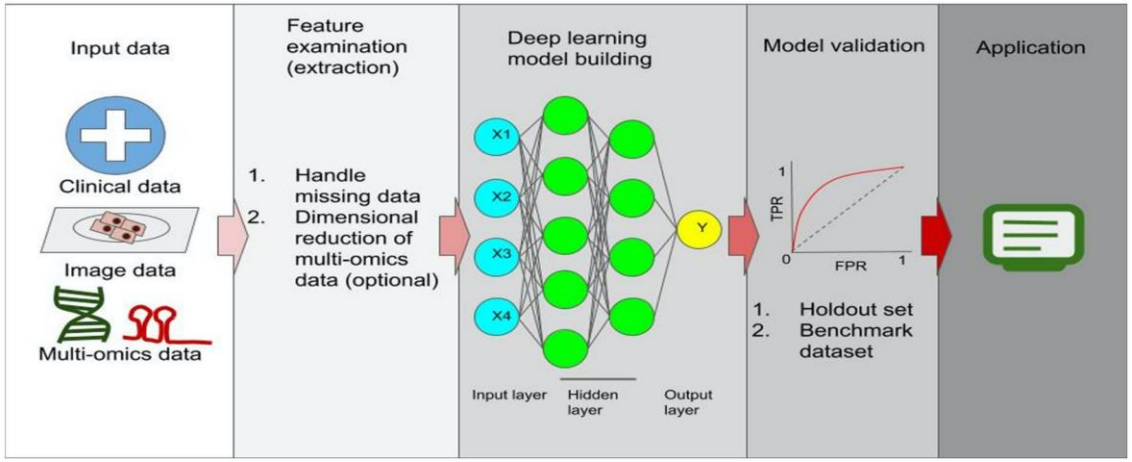
- Screening
- Diagnosis
- Therapy
- **Prognosis**
- …

Review

The Application of Deep Learning in Cancer Prognosis Prediction

Wan Zhu [1,2,*,†], Longxiang Xie [1,†], Jianye Han [3] and Xiangqian Guo [1,*]

[1] Department of Preventive Medicine, Institute of Biomedical Informatics, Cell Signal Transduction Laboratory, Bioinformatics center, School of Basic Medical Sciences, Henan University, Kaifeng 475004, China; xielongxiang123@126.com
[2] Department of Anesthesia, Stanford University, 300 Pasteur Drive, Stanford, CA 94305, USA
[3] Department of Computer Science, University of Illinois, Urbana Champions, IL 61820, USA; mr.neohan@gmail.com
* Correspondence: wanzhu@stanford.edu (W.Z.); xqguo@henu.edu.cn (X.G.)
† These authors contributed equally to this work.



Benefits of Machine Learning

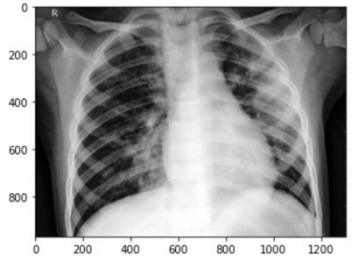
- Efficient screening tools that reduce demand on costly health care resources
- Exploitation of available data about patients by data fusion from multiple sensors
- Help physicians cope with the information overload
- Optimize allocation of hospital resources
- …

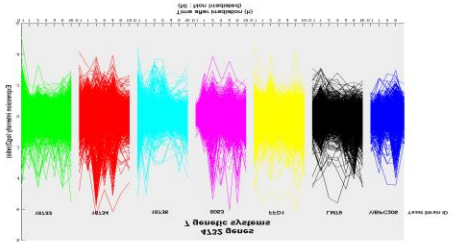
Different types of data (ajouter une image de notes)

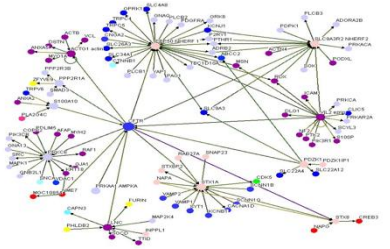
- **Text data: Medical notes**
- Tabular data: continuous, categorical, and ordinal
- Image data
- Time series / sequential data
- Graphs
- …

<B>ASSESSMENT AND PLAN: </B>This is a 43-year-old gentleman who underwent what sounds like a sigmoid colectomy with end colostomy and Hartmann's pouch in November of 2007 secondary to perforated colon from diverticulitis. The patient presents for reversal of his colostomy as well as repair of his incisional hernia. I have asked the patient to return to High Point Regional and get his medical records including the operative note and pathology results from his initial surgery so that I would have a better idea of what was done during his initial surgery. He stated that he would try and do this and bring the records to our clinic on his next appointment. I have also set him up for a barium enema to study the rectal stump. He will return to us in two weeks at which time we will review his radiological studies and his medical records from the outside hospital and determine the best course of action from that point. This was discussed with the patient as well as his sister and significant other in the clinic today. They were in agreement with this plan. We also called the social worker to come and help the patient get more ostomy appliances, as he stated that he had no more and he was having to reuse the existing ostomy bag. To my understanding, his social worker, as well as the ostomy nurses were able to get him some assistance with this.

Cell Type	S100A8	FCER1A	FCGR3A	GNLY	PPBP	MS4A1	KLRB1	LGALS3	CST3	CD79A	NKG7
CD4 T	-0.426564	-0.068512	-0.351253	-0.260893	-0.098842	-0.320419	-0.024082	-0.355035	-0.536417	-0.331537	-0.397536
CD14 Monocytes	1.941531	-0.096441	-0.310415	-0.230350	0.030063	-0.323063	-0.211329	1.211140	1.599805	-0.353766	-0.387982
B	-0.465240	-0.033093	-0.306757	-0.331799	-0.107808	2.065213	-0.252235	-0.377599	-0.446173	2.269075	-0.533727
CD8 T	-0.521894	-0.020857	-0.008047	0.161251	-0.106239	-0.342383	0.429695	-0.216598	-0.544552	-0.410379	1.335278
NK	0.186811	-0.368140	1.816293	-0.165194	0.223581	-0.163347	-0.205593	0.609435	1.234201	-0.226449	-0.138674







Medical notes classification

Objective: Classifying Medical notes to the medical specialty based on a variant of BERT models.

Dataset Context: Kaggle Medical Notes 2019

Input: Free-text medical notes available at: LINK

Content:

The filenames are numbered from 1001 to 2239, where:

Train data: 1001.txt to 1826.txt

trainLabels.csv - the labels for training data

Output: Medical specialty (Orthopedic, Radiology, Urology, Gastroenterology, ...)

What is a medical note?

- documents a patient's healthcare visit and comprises part of his or her secure electronic health record (EHR) chart.
- are written by physicians or any other member of the medical team following an office visit, a telemedicine visit, a procedure, testing, therapy, …
- ensures systematic documentation of a patient's medical history, diagnoses, past and current medications, allergies, treatment, and overall care.

Practical course (project) Goal

- Build a medical text classification pipeline
- Implement a TF-IDF + SVM baseline
- Fine-tune BERT (or its variant) for clinical text classification
- Compare performance, cost, and practicality of classical vs transformer models
- Understand when BERT is suitable in medical NLP
- Critical thinking about the huge models based on transformers

Medical notes classification

Part 1

- Load the dataset and identify the classes

Analyse:

Note length distribution

Label distribution

Content of the notes

- Apply relevant NLP processing steps on the dataset
- Use a simple tf-idf tokenizer and an SVM to classify the medical notes
- Use the Bert tokenizer and fine - tuned Bert model to classify the notes.

You can rely the code provided in LINK a starting point and the code available on for finetuning BERT LINK1

- Evaluate using suitable metrics, analyze, compare, and discuss the results (Does BERT improve the results for all classes? Where does it fail? Is improvement worth the cost?)

Medical notes classification

Part 2

- Cite the advantages and Risks in medical NLP: Interpretability, Overfitting, Deployment cost, Energy consumption, …

Key discussion points:

  - Needs for Healthcare
  - Risks of using AI in general and huge Transformers based models in particular

- What are the alternatives to using BRT?

Medical notes classification

Part3 (Optional)

- Implement the alternative proposed in part 2 and discuss the obtained results

Remark : If **part 2** is performed before **part 1**, you can directly replace BERT by its alternative (no need to finetune BERT)